

Research Statement

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We live in a digital era that increasingly relies on distributed systems that operate across heterogeneous environments, participants, and trust assumptions. Emerging decentralized infrastructures promise new forms of transparency, collaboration, and applications. However, their growing complexity presents fundamental challenges to security, privacy, scalability, and incentive compatibility. Addressing these challenges requires new cryptographic techniques and systems foundations that account for the realities of modern distributed computing.

My research lies at the intersection of cryptography, security, privacy, and decentralized systems. The central question that drives my work is: **How can we build secure, privacy-preserving, and scalable decentralized systems under heterogeneous trust, participation, and incentive models?** A common theme across my efforts to address this question is bridging theory and practice; I design secure and efficient systems and protocols that connect rigorous cryptographic foundations with real-world applications. My work spans distributed cryptographic protocols, privacy-preserving computing, and decentralized (blockchain-based) systems. These efforts have produced both foundational advances and practical impact. My PhD thesis work, a system called CacheCash, led to a startup to promote its practical deployment. My subsequent work has attracted support from NSF, Protocol Labs, Uniswap Foundation, and DHS, as well as recognition through fellowships, awards, invited talks, and collaborations with industry.

My current research activities are organized around three interconnected thrusts: secure and scalable decentralized systems, privacy-preserving computing, and cryptographic foundations for physical hardness assumptions. Together, these thrusts aim to establish the foundations of trustworthy and performant decentralized systems and distributed cryptographic protocols that capture real-world settings.

Current Research Activities

Secure and Scalable Decentralized Systems. A significant portion of my research focuses on blockchain-based systems. The unique features of this technology—transparency, public verifiability, and decentralization—offer new paradigms for reshaping the Internet and its digital services (what is now called Web 3.0). These systems, however, suffer from serious scalability issues and introduce new attack vectors due to their open-access and monetary-incentivized setting. My research seeks to address these challenges so we can enjoy the benefits of blockchains without degrading performance or empowering attackers.

My work on Web 3.0 started with my PhD thesis [1], where we built CacheCash, a decentralized content distribution system that allows end users to organically set up new caches and distribute content in exchange for cryptocurrency payments. To address the challenges encountered as the system evolved, we developed several components related to threat modeling, cache accounting attacks, and decentralized probabilistic micropayments [5, 13, 6]. After joining UConn, I expanded this agenda to address broader questions surrounding the scalability, security, privacy and economic aspects of Web 3.0 (I will discuss privacy separately).

On the *secure scalability* side, I study decentralized resource markets [2], in which a blockchain infrastructure is used to facilitate exchange of digital resources in a peer-to-peer fashion. We developed two scalability solutions, chainBoost [23] and chainScale [24], which build a new sidechain architecture that enables workload sharing and blockchain pruning. These systems significantly improve throughput, latency and storage overhead of resource markets while preserving their security guarantees. I extended these ideas to decentralized finance (DeFi) through ammBoost [21] that aims to solve the scalability issues of automated market makers (AMMs). Building on these

efforts, I am currently developing modular approaches (by utilizing the MoSS framework [20]) for formalizing and proving security of Web 3.0 systems under realistic settings. I believe this work will promote formal security analysis of large-scale distributed systems.

As blockchains operate in economically-motivated settings, new attack classes emerge from strategic behavior rather than purely technical vulnerabilities. Thus, it is important to study these *economic aspects and their security implications*. My work has examined topics related to cryptocurrency wallets, collateral management, and incentivized attacks. In [14], we investigated the notion of timed, anonymous, and revocable delegation of digital signatures, which enables controlled delegation of cryptocurrency accounts and wallets. We also studied collateral management in wallets (and layer-2 protocols in general) [10]. We developed several competitive policies under various settings that improve how collateral can be managed when backing-up on-chain activities.

In terms of incentivized attacks, I studied the notion of criminal smart contracts (CSCs), in which smart contracts and blockchain oracle services are used to orchestrate collaborative attacks against real-world targets in a decentralized way. We showed the feasibility of trustless, collaborative attacks against targets residing in the real world [22]. Currently, I am investigating more sophisticated incentive-driven CSC-based attacks, and the design of corresponding defensive frameworks. More recently, I started investigating selfish mining attacks in Bitcoin, which allow miners to collect more rewards than their fair share. We have studied temporal profitability and adaptive selfish mining strategies while accounting for various reward/fee models [17, 18]. These lines of research shed light on misuses of blockchain systems and the impact of incentives, providing valuable insights into long-term Web 3.0 security.

Privacy-preserving computing. Privacy-preserving computing enables mutually-distrusted parties to collaborate and compute over their private data without disclosing this data. My work in this area focuses on devising new protocols and applying existing ones to enable meaningful computation without sacrificing privacy. These efforts mainly cover multiparty computation (MPC) protocols and fully homomorphic encryption (FHE).

I began working on MPC during my PhD studies through projects involving private genome analysis [27], and compiler support for private data structures [28]. Later on, and inspired by recent trends on utilizing the blockchain model in cryptography, I took these efforts further to develop new models for MPC. In Gage MPC [4], we devised a paradigm that incorporates miners as active participants in the private computation. Gage MPC guarantees fairness and output delivery, while circumventing the lower bound of leaking the residual function in non-interactive MPC. More recently, I have explored MPC in settings where participants have unequal resources and trust levels. We developed weighted MPC protocols that support computing parties with weights in both static and dynamic (fluid) settings [12]. These protocols allow security assumptions to be expressed in terms of party weights rather than the number of parties, bringing secure computation closer to the realities of decentralized systems (as in proof-of-stake consensus).

FHE provides an alternative approach for secure computation by computing directly over encrypted data. FHE offers strong privacy guarantees and reduces interaction during the computation, making it attractive for Web 3.0 applications. Indeed, privacy is a huge concern in blockchain systems given that most permissionless blockchains log their content in the clear. To address this issue, we conducted a thorough review of existing blockchain privacy-preserving solutions in [15]. We also designed smartFHE [25], a system that utilizes FHE and zero knowledge proofs to support privacy-preserving smart contracts. These efforts motivated us to optimize efficiency and usability of programming with FHE. We built Parasol [26], an end-to-end compiler that enables efficient FHE circuit composition to extract parallelism, while providing a simple interface for developers. Currently, we are exploring the utility of scheme switching to further optimize FHE programs, as well as various applications in distributed settings.

Cryptographic foundations for new trust/hardness models. Another line of my research explores how physical assumptions, namely, hardware tokens, can expand the capabilities of cryptographic systems. These enable circumventing impossibility results, improving efficiency, or building new applications that are impossible using software alone. In [8], my collaborators and I (a team of cryptographers and biologists) developed a mechanism for generating and manipulating protein polymers to obtain a new type of consumable storage that exhibits self-destruct properties, assuming the hardness of certain polymer-sequencing problems. We developed a formalism that captures the functionality and security properties provided by the technology. Then, using this technology, we constructed cryptographic applications, including a password-controlled, self-destructible secure storage, and (a relaxed version of) one-time programs that allow evaluating a secret function only a limited number of times. Later on, we investigated the relation between the unclonability offered by these biological tokens and quantum computing [9]. Building on these ideas, I showed how to build password-authenticated cryptography, including MPC and delegation of cryptographic capabilities, from consumable storage tokens [3]. These efforts open several future research directions highlighting the benefit of investigating alternative hardness assumptions for cryptography.

Future Research Directions

Looking forward, my goal is to develop the foundations of trustworthy distributed computing in environments characterized by heterogeneous trust assumptions, dynamic participation, and increasingly complex interactions within emerging decentralized infrastructures.

Security and privacy of heterogeneous systems and computing models. A primary future direction of my research is understanding the security and privacy of heterogeneous computing models. These include discrepancies within the same system or across several systems interacting with each other. For example, how to accommodate participants with different computation capabilities and trust levels, systems with different security and privacy requirements, or even hybrid systems that may have legacy services coexisting with new decentralized (e.g., blockchain-based) ones. Such discrepancies present various security and privacy challenges, which require new paradigms and protocols to address them.

Under this direction, my future research plans include investigating Web 3.0 systems that interact with the real world and its legacy systems. Examples include identity management that requires attestation from governmental agencies, supply chain management of physical products, or those that need legal intervention to enforce compliance. As a preliminary step, we have conducted a thorough systematization study of blockchain oracles [16], i.e., services that provide real-world data feeds to Web 3.0 applications, delineating their current capabilities and limitations. Building on that, I plan to develop a generic framework that enables a more expressive interaction with the real world and its legacy services, while preserving privacy and security, and ensuring compliance, across such diverse domains. I also plan to expand my work on MPC in heterogeneous settings to support open dynamic models (e.g., the you speak only once model—YOSO [19]), and explore their utility for Web 3.0-related applications.

Connections to AI and machine learning. The rapid proliferation of AI agents has motivated the community to explore a bidirectional relation with security and privacy. That is, build trustworthy machine learning and agentic AI systems, while also examining the utility of these powerful agents in reasoning about security and building defenses. Some of my recent works fall under this category. These include addressing participation anonymity and privacy in federated learning [11], and studying the security of learned data structures, namely learned Bloom filters [7], in which machine learning models are utilized to improve accuracy and efficiency of such structures.

Expanding on these efforts, my future research plans include utilizing AI agents for automating security auditing of cross-chain bridges and interoperability solutions; connecting AI agents with

blockchains in a secure and scalable way while studying the implications of such combination; as well as investigating the utility of learned data structures in the design of scalable and secure decentralized systems.

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